

1. a) Use $\mathbf{A} = \frac{\mu I l e^{-jkr}}{4\pi r} (\cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\boldsymbol{\theta}})$ in $\mathbf{E} = -\frac{j\omega}{k^2} (k^2 + \nabla \nabla \cdot) \mathbf{A}$ to get the expression

$$\mathbf{E} = -\frac{I \ell k^2}{4\pi} \eta \left\{ \left[\frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] 2 \cos \theta e^{-jkr} \hat{\mathbf{r}} + \left[\frac{1}{jkr} + \frac{1}{(jkr)^2} + \frac{1}{(jkr)^3} \right] \sin \theta e^{-jkr} \hat{\boldsymbol{\theta}} \right\}$$

Throughout your calculation underline all terms associated with the scalar potential, which are all terms that come from $\nabla \nabla \cdot \mathbf{A}$.

- (b) You will read in some books that the $\nabla \nabla \cdot \mathbf{A}$ calculation is not necessary. State specifically where in your part (a) calculation the far field would change if $\nabla \nabla \cdot \mathbf{A}$ were not in

$$\mathbf{E} = -\frac{j\omega}{k^2} (k^2 + \nabla \nabla \cdot) \mathbf{A}.$$

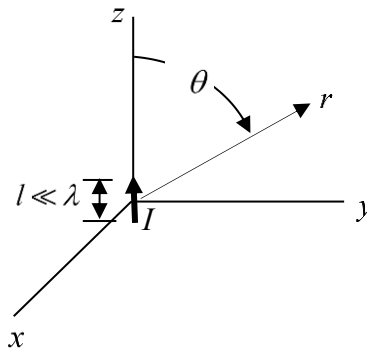
2. Using the method of gradients :

- (a) find $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ and $\hat{\mathbf{z}}$ in terms of spherical coordinates r, θ, ϕ and unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$
 (b) find $\hat{\boldsymbol{\phi}}$ in terms of spherical coordinates r, θ, ϕ and Cartesian unit vectors.

3. For a short dipole, where $l \ll \lambda / 2$ and a realistic current distribution is $I = I_0 [1 - \frac{2|z|}{l}]$, find the

following showing all calculations and compare each result with the Hertzian dipole result obtained in the notes.

- (a) Far electric field
 (b) Time average power density
 (c) Radiated power
 (d) Radiation resistance
 (e) Directivity
 (f) Gain



4. A small circular loop of current lies in the x-y plane. The current has a constant amplitude I and the radius of the loop is $a \ll \lambda$.

- (a) Show that $|\mathbf{r} - \mathbf{r}'| \cong r - a \sin \theta \cos(\phi - \phi')$
 (b) Using the approximation in (a) for the phase term and $|\mathbf{r} - \mathbf{r}'| \cong r$ for the amplitude term, find the electric vector potential in the far field. When carrying out your calculation show that the integration result contains a J_1 Bessel function. Then show that the argument is small and use the Bessel function small argument approximation $J_1(x) \cong x/2$ to obtain your final result. Reference equation numbers from Abramowitz and Stegun for the integration and for the small argument approximations you use in your solution. Hints: see Notes 6 example and take care concerning the unit vector for the current – the unit vector $\hat{\boldsymbol{\phi}}'$ in your expression for the current changes its direction as you integrate around the loop, so it must be expressed in terms of the fixed observation point coordinates r, θ , and ϕ which have fixed unit vectors $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}$, and $\hat{\boldsymbol{\phi}}$.
 (c) Write down the far electric field using the far field approximation.

